Agentic Compliance Auditor

A plain-English tour of the frontend we built (Day 1 and Day 2)

This document walks through everything we've built for the user-facing part of the Agentic Compliance Auditor so far. It is written for a non-technical reader — no code, no jargon. If you want to understand what the screen does, why we made each choice, and where this is heading, this is your guide.

> *Big picture: We are building a specialised assistant that reads compliance documents (GDPR, HIPAA, SOC2) and answers questions about them. Unlike a chatbot, it also* ***shows you the exact passage*** *it used to answer, highlighted right on the original page.*

1. What problem is this solving?

Compliance teams at companies (banks, hospitals, software firms) spend hours hunting through long regulatory documents to answer questions like:

- "Do our data retention rules meet GDPR Article 17?"
- "What exactly does HIPAA require when patient data is accessed?"
- "What controls does SOC2 want for user access management?"

A normal chatbot like ChatGPT can attempt these questions, but it has two big problems in a compliance setting:

- It sometimes **invents** details that sound plausible but aren't in the law.
- It rarely **cites** its source, so a compliance officer can't verify the answer.

Our tool fixes both. It only answers using words that are actually in the documents, it always tells you which document, which page, and which paragraph — and it lets you click the citation to see it lit up on the page in front of you.

2. What you see when you open the app

The app is a single screen split into two halves.

Left half — the document viewer

This is where the actual PDF of the regulation is shown. Right now the app comes with three sample PDFs — GDPR, HIPAA, and SOC2 — and you switch between them using the tabs in the top-right corner. Above the PDF is a small strip showing which file is open and which page you're on. Below the strip are Next/Previous buttons for turning pages manually.

> *Think of the left half like the physical book sitting on the desk of a compliance officer. It's the source of truth. Nothing the AI says is more trusted than what's written here.*

Right half — the workspace

This is where you interact with the AI. It has four stacked areas:

- **Question box** at the top — type your question, press Enter or click Ask.
- **Agent trace** in the middle — a live checklist of what the AI is doing right now.
- **Answer** below that — the AI's reply with small clickable citation chips.
- **Sources** at the bottom — the list of document passages the AI actually read.

3. The Agent Trace — why this isn't an ordinary chatbot

Most AI chatbots feel like a magic box: you ask something, a spinner spins, an answer appears. You have no idea what happened in between, so if the answer is wrong you can't tell why. In our app we deliberately open that box up.

When you click Ask, three tick-boxes appear on the right side and light up in sequence:

- **Retrieving chunks** — the AI is scanning the regulation to find relevant passages. When done, it shows a count (e.g. "3 chunks").
- **Generating draft** — the AI is writing a first-attempt answer using those passages.
- **Reflecting on answer** — the AI is checking its own work. If it thinks the answer is weak or unsupported, it can go back and try again.

> *This third step is the special sauce. Most AI systems never look at their own answer before sending it. Ours does. The technical term is **self-reflection**, and it's the difference between an AI that confidently gives you a wrong answer, and one that notices it's not sure and tries harder.*

For a compliance officer, this visible workflow is reassuring. They can see the AI did search real documents, they can see it double-checked its answer, and if something looks off, they know exactly which step to investigate.

4. Clicking a citation — the demo moment

Once the answer appears, look carefully at the sentences. You'll see little blue chips scattered through the text — things like **[1] p.2** or **[3] p.4**. These are the citations. Each one says "this claim comes from page X of the document."

Here's the important part: **the chips are buttons**. When you click one:

- The left-hand PDF instantly jumps to that page.
- The exact sentences the AI relied on get **highlighted in yellow**.
- Your eye can go straight from the AI's claim to the original wording — no scrolling, no searching.

This is the moment we're building for the LinkedIn screenshot. It shows off in one glance the two things that make this project distinctive: ***the AI shows its work****, and* ***every claim is one click away from the source****.*

5. The regulation tabs (GDPR, HIPAA, SOC2)

Compliance rarely lives in one book. A hospital that also runs a European service has to comply with HIPAA in the US and GDPR in the EU. A cloud provider selling to enterprise adds SOC2 on top. Our app treats each of these as its own "lane":

- Clicking a tab loads the correct PDF into the viewer.
- It also tells the AI to focus its search on that regulation only.
- The workflow trace and answer refresh accordingly.

In future versions, this is where the app will grow into a genuine multi-agent system — where separate specialist agents (one for GDPR, one for HIPAA, one for SOC2) work in parallel on questions that span more than one regulation. Today's tabs are the visual foundation for that.

6. What's real today, and what's a stand-in

We built the frontend using a technique called **mocking** — meaning we hand-crafted a fake response so we could develop and test the user interface before the AI backend was ready. Here's the split:

Piece	Status	What that means
Split-screen layout	Real	This is the final design.
PDF viewer	Real	Uses your actual GDPR, HIPAA, SOC2 PDFs.
Regulation tabs	Real	Switches PDFs live.
Question box & Ask button	Real	Fully working input.
Workflow trace animation	Real	Ticks light up in order, exactly like production will.
The AI's answer text	Placeholder	A pre-written sample answer, not from a live AI yet.
Citation chips + jump-to-page	Real	Actually finds and highlights the passage in the PDF.
Yellow highlighting on the page	Real	Works on real PDF text, not a fake overlay.

> *The reason the answer text is still a placeholder: your local machine cannot handle the heavy AI processing needed to answer live questions (it ran out of memory when we tried). The plan is to do that one heavy setup step on a free cloud service later, then plug it in. Everything the frontend needs to display is already wired to receive real answers the moment they're available.*

7. Why we made the choices we did

Why not just extend the old Streamlit page?

Streamlit is quick to build but limited — it's essentially a form-and-output tool. You cannot cleanly build a two-panel layout with a real PDF viewer, live-updating workflow ticks, and interactive citation chips inside it. Moving to React (a modern web framework used by companies like Facebook, Netflix, and Airbnb) gives us the flexibility for all of that **and** makes the project look far more professional on your GitHub.

Why show the workflow trace at all?

Trust. In a compliance context, an answer without visible reasoning is close to useless. The trace turns "here's an answer, trust me" into "here's what I did, here's why I think this is right, and here's the source." That's a big difference in a regulated environment.

Why yellow highlights on the actual PDF instead of just quoting the text?

A quote in a chat window is easy to misread out of context. A yellow line on the original page instantly tells the reader "here's the paragraph, here's what surrounds it, judge for yourself." It also demonstrates something impressive technically that recruiters and hiring managers will notice: the app is deeply connected to the source document, not just skimming it.

Why start with mocked data?

Two reasons. First, it lets us verify the design and feel of the app before spending time on the heavy AI plumbing. Second, and more importantly for you: it means the frontend is already fully working and ready to show, even before the backend can serve live answers. You could put screenshots on LinkedIn today.

8. What's next

There are three more short pieces of work to make this a complete portfolio piece:

- **Cloud data setup (one-time)** — do the heavy AI preparation step for free on Google Colab, download the resulting index, and plug it into the app so live answers work.
- **Real answers wired in** — swap the placeholder answers for live ones. Everything on the frontend is already ready for this; it's a one-line switch.

- **Polish for the demo** — small quality-of-life touches like a loading shimmer on the PDF, an error message if something breaks, and a short (10-second) screen recording for your LinkedIn post.

The takeaway

In two days of focused work you now have the visible, screenshot-able heart of the project working: a two-panel compliance auditor with a live agent trace and click-to-source citations. Anyone who sees this on your GitHub or LinkedIn will immediately understand that you're not shipping just another wrapper around ChatGPT — you're shipping a specialised AI product with a real user experience.